Venkata Lakshmi Parimala Pasupuleti

Senior Software Engineer · Python · Bash · SQL · Object-Oriented Programming · REST APIs · CI/CD Automation · Deployment Scripting · Observability · Telemetry · RCA · High Availability · AWS

+1-703-261-5410 | venkatalakshmiparimala@gmail.com | linkedin.com/in/venkata-lakshmi-parimala-pasupuleti | github.com/venkatalakshmiparimala/AgentProbe |
Virginia, USA — Open to Relocation (Sunnyvale CA)

PROFESSIONAL SUMMARY

Software engineer with 8 years building and maintaining **large-scale enterprise backend systems** in Python, Bash, and SQL - owning the full lifecycle from design through deployment, monitoring, and RCA. At Entrata: **Python and Bash automation pipelines** eliminating 80% of manual operational workload; **200+ SQL query optimizations** (30% performance gain) across PostgreSQL and MySQL via execution plan tuning, indexing, partitioning, and audit tables; **RabbitMQ async processing** with retry logic and idempotency guarantees; **REST API design and service-based integrations** across 20+ production modules serving 100K+ MAU at 99.5% uptime; AWS S3 pipelines with 100% PHI compliance; **internal debugging and telemetry tools** reducing debugging time 40% and keeping open bugs below 10 for 6+ months. Built **AgentProbe**: full observability platform with CI/CD automation scripts, regression gate, performance metric tracking, and deployment validation across environments. M.S. Data Analytics Engineering, George Mason University (GPA 3.87).

WORK EXPERIENCE

George Mason University
Aug 2024 – May 2026

Research Associate / Graduate Teaching Assistant — Python · Bash · SQL · Statistical Modeling · Observability

- Built multi-source Python data pipelines (190+ countries, WHO + UK Road Safety datasets) achieving **$R^2 = 0.977$**; applied Isolation Forest for automated data integrity validation across 3 research pipelines; produced Tableau and ggplot2 observability dashboards improving comprehension 40%. Delivered Python and database design workshops to **600+ students**.

Entrata — Enterprise PropTech SaaS
Sep 2018 – Aug 2024 · Remote

Senior Software Engineer — Python · Bash · SQL · REST APIs · OOP · CI/CD · AWS · High Availability · RCA

- **Python and Bash automation** eliminated 80% of manual operational workload - pipelines transforming federal HUD source rows into ~180K domain records powering eligibility and compliance decisions; **CI/CD scripting** (GitHub Actions) with pre and post validation gates; **deployment automation** across multi-tenant environments; all scripts maintained through production incidents and annual release cycles.
- **Optimized 200+ SQL queries** (30% performance gain) across PostgreSQL and MySQL via execution plan analysis, indexing, partitioning, stored procedures, triggers, and audit tables; owned **schema design and normalization** for 20+ production modules; **RabbitMQ async processing** with retry logic and idempotency guarantees; **Redis caching** layer for high-frequency data access paths, reducing database load on hot queries.
- Built **internal telemetry and observability tooling** - debug dashboards reducing debugging time 40%, bug tracking keeping open defects below 10 for 6+ months; conducted **RCA analysis** on production incidents with corrective documentation; performed corrective and adaptive **maintenance** procedures; monitored application performance and availability across 99.5% uptime SLA; led PHI-compliant **AWS S3 migration** (Libsodium, zero security incidents); **300+ code reviews; mentored 5+ engineers**.

Entrata India (Db Xento Systems)
Jan 2017 – Sep 2018 · Pune

Associate / Trainee Software Engineer — Python · SQL · Bash · REST APIs · AWS

- Built Python and SQL backend modules for property management and compliance; wrote Bash deployment scripts and post-validation checks; led AWS S3 cloud migration; designed SQL import pipelines processing 1.6M federal housing records; contributed to RCA and defect analysis under senior engineer guidance.

PROJECTS

AgentProbe: CI/CD Automation, Observability & Deployment Validation Platform
OBSERVABILITY · CI/CD

Python · Bash · FastAPI · Docker · CI/CD (GitHub Actions) · ChromaDB · REST APIs · React — github.com/venkatalakshmiparimala/AgentProbe

- **CI/CD automation scripts** with pre and post validation gates blocking quality regressions before merge; **telemetry and observability dashboard** (React) tracking per-run performance metrics, latency trends, and retrieval drift; 15-endpoint FastAPI backend with REST API service integrations; **deployment validation** across environments with structured performance logging; validated 1,240+ test cases with automated regression testing: pass@1 81.7%, pass@3 88.3%.

Leasel: Enterprise Backend Platform — Python OOP · REST APIs · High Availability
PRODUCTION BACKEND

Python · FastAPI · LangGraph · SQLite · REST APIs · Docker · Streamlit — github.com/venkatalakshmiparimala/Lease_complaince_rag

- Object-oriented Python backend with **8-component modular architecture**, FastAPI REST service layer, SQLite persistence, Docker containerization, and Streamlit engineer console; **high-availability design**: confidence-scored outputs, deterministic fallbacks, and HITL escalation prevent silent failures; structured logging and performance tracking for every request path. Results: 93.6% precision, 0% unsupported-claim rate on 1,240 compliance documents.

AI on FHIR: REST API Integration & Backend Automation
REST APIS · BACKEND

Python · Flask · REST APIs (FHIR) · Docker · spaCy · BERT · JavaScript · React — github.com/venkatalakshmiparimala/AI_on_FHIR

- Python Flask backend with **REST API service integration** (FHIR standard) - OOP architecture translating natural-language queries into structured API calls; HIPAA-compliant RBAC; Docker containerized deployment; ~60% reduction in manual retrieval time. Demonstrates **end-to-end backend automation across a third-party REST API**.

EDUCATION

M.S. Data Analytics Engineering — George Mason University
May 2026 · GPA 3.87

Software systems, data engineering, statistical modeling, ML pipelines, performance engineering, observability

B.Tech Computer Science & Engineering — JNTU Anantapur
2013 – 2017

TECHNICAL SKILLS

Backend & Automation: Python (expert), Bash/Shell scripting, SQL (expert — 200+ production optimizations, execution plan tuning, indexing, partitioning, stored procedures, triggers, audit tables), PL/SQL, PHP, JavaScript, TypeScript, R; Object-Oriented Programming (OOP), REST APIs, Service-Based Integrations, RabbitMQ async messaging, Microservices, Docker, CI/CD (GitHub Actions), Deployment Automation, Pre/Post Validation Scripts, Telemetry, Observability, Performance Monitoring, RCA Documentation, High Availability, High Scalability, Git, Linux, Agile/Scrum

Databases & Cloud: PostgreSQL (expert), MySQL, Oracle, Redis (caching), NoSQL (SQLite, ChromaDB/vector DBs); AWS (S3, EC2, RDS, Glue), Libsodium encryption, PHI compliance; 300+ code reviews, 5+ engineers mentored

AI & ML Integration: Python ML pipelines (PyTorch, scikit-learn, XGBoost, Isolation Forest), LangGraph, LangChain, RAG Pipelines, LLM Evaluation, FastAPI, Prompt Engineering, NLP (spaCy, BERT), Tableau, Power BI, Statistical Modeling